

Virtual Cybersecurity Lab: Network Security Monitoring, IDS and Firewall Configuration

Practical work with VirtualBox, Kali Linux, Metasploitable 2/3, Snort and pfSense

Scope: Built an isolated virtual cybersecurity lab, established connectivity between vulnerable systems, deployed Snort for custom and built-in intrusion detection, and configured pfSense for routing, firewall filtering, and port forwarding.

- Virtual Network Configuration

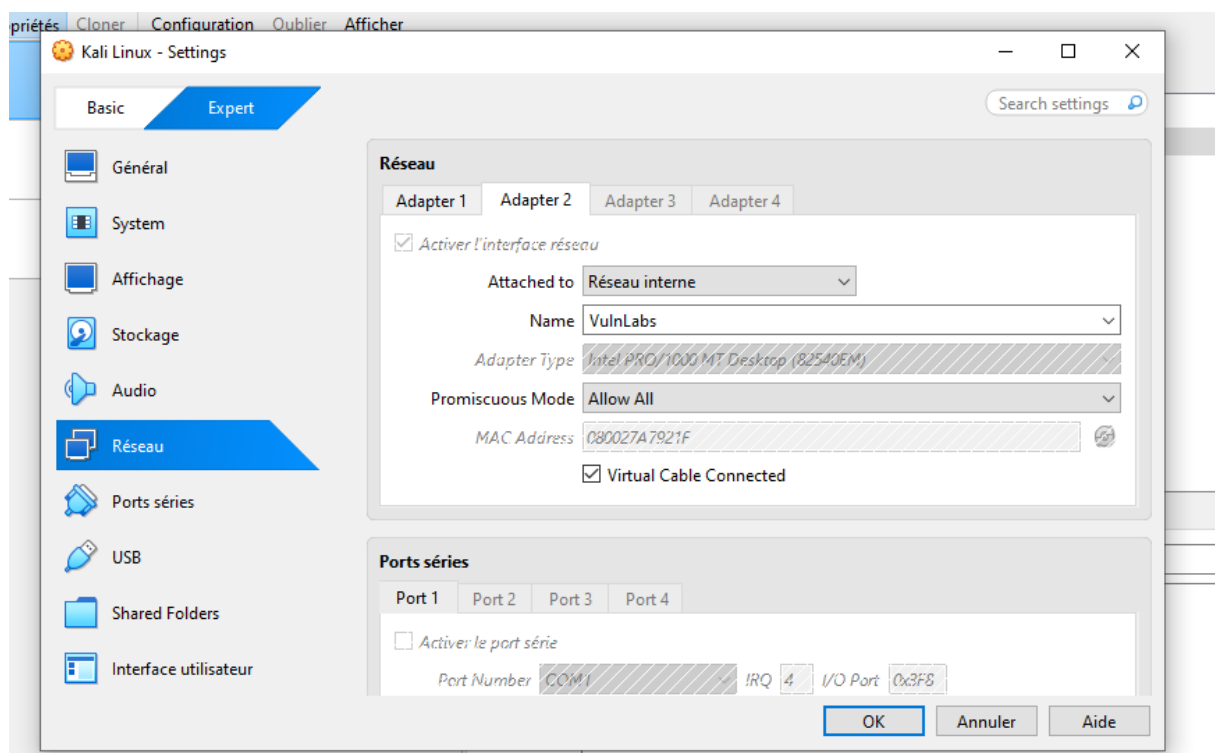
Configured the VirtualBox DHCP service for the VulnLabs internal network, including the lab subnet and IP address range used by the virtual machines.

```
C:\Program Files\Oracle\VirtualBox>vboxmanage dhcpserver add --network=VulnLabs --server-ip=172.30.1.1 --lower-ip=172.30.1.20 --upper-ip=172.30.1.50 --netmask=255.255.255.0 --enable

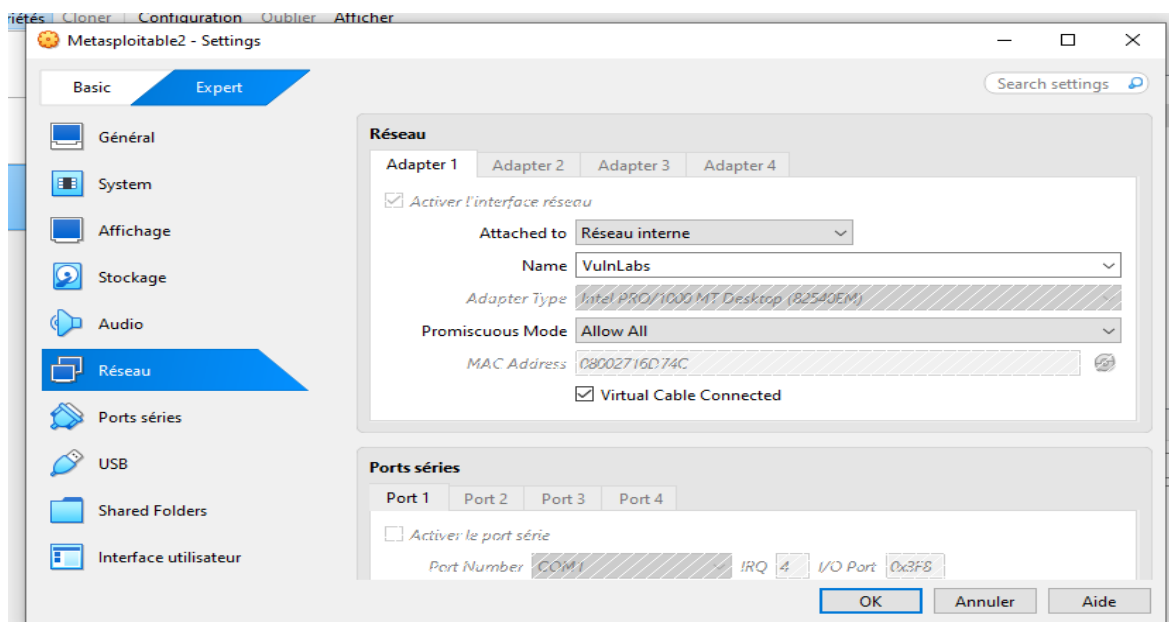
C:\Program Files\Oracle\VirtualBox>vboxmanage dhcpserver add --network=HiddenVulnLabs --server-ip=10.205.1.1 --lower-ip=10.205.1.20 --upper-ip=10.205.1.50 --netmask=255.255.255.0 --enable

C:\Program Files\Oracle\VirtualBox>
```

- Configured the Kali Linux VM in VirtualBox to use the VulnLabs internal network, allowing it to communicate with the vulnerable lab machines.



- Configured Metasploitable 2 in VirtualBox to use the same VulnLabs internal network as Kali Linux.



- Verified connectivity between Kali Linux (172.30.1.20) and Metasploitable 2 (172.30.1.21) using ICMP ping. Successful replies confirmed that both VMs were correctly connected to the same lab network.
- Performed service and version detection against Metasploitable 2 using Nmap (-sV) to identify open TCP ports and the services and versions exposed on the target.

```

kali@kali:~$ nmap -sV 172.30.1.21
Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-29 07:40 -0400
Nmap scan report for 172.30.1.21
Host is up (0.00042s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
22/tcp    open  ssh          OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp    open  telnetd     Linux telnetd
25/tcp    open  smtp        Postfix smtpd
53/tcp    open  domain      ISC BIND 9.4.2
80/tcp    open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp   open  rpcbind     2 (RPC #100000)
139/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp   open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp   open  exec        netkit-rsh rshd
513/tcp   open  login       OpenBSD or Solaris rlogind
514/tcp   open  shell       Netkit rshd
1099/tcp  open  java-rmi    GNU Classpath grmiregistry
1524/tcp  open  bindshell   Metasploitable root shell
2049/tcp  open  nfs         2-4 (RPC #100003)
2121/tcp  open  ftp         ProFTPD 1.3.1
3306/tcp  open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp  open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp  open  vnc         VNC (protocol 3.3)
6000/tcp  open  X11         (access denied)
6667/tcp  open  irc         UnrealIRCd
8009/tcp  open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp  open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 08:00:27:16:D7:4C (Oracle VirtualBox virtual NIC)
Service Info: Hosts: metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

```

- Deployed Metasploitable 3 as a vulnerable Windows VM using Vagrant and VirtualBox.

```

C:\Users\hp>cd .\vagrant.d\boxes
C:\Users\hp\.\vagrant.d\boxes>dir
Le volume dans le lecteur C n'a pas de nom.
Le numéro de série du volume est C03E-A557

Répertoire de C:\Users\hp\.\vagrant.d\boxes
29/08/2026 13:58 <DIR> .
29/08/2026 13:58 <DIR> ..
29/08/2026 13:58 <DIR> rapid7-VAGRANTSLASH-metasploitable3-win2k8
0 fichier(s) 0 octets
3 Rép(s) 111 625 408 512 octets libres

C:\Users\hp\.\vagrant.d\boxes>REN "rapid7-VAGRANTSLASH-metasploitable3-win2k8" "metasploitable3-win2k8"
C:\Users\hp\.\vagrant.d\boxes>dir
Le volume dans le lecteur C n'a pas de nom.
Le numéro de série du volume est C03E-A557

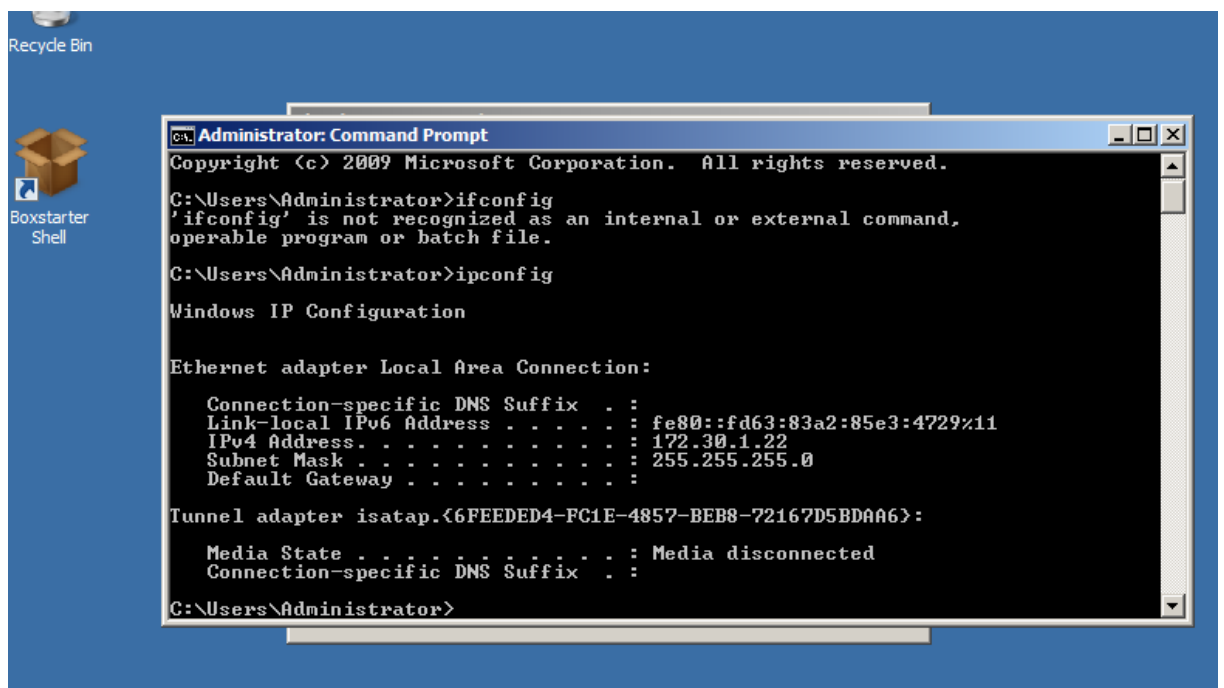
Répertoire de C:\Users\hp\.\vagrant.d\boxes
29/08/2026 14:24 <DIR> .
29/08/2026 14:24 <DIR> ..
29/08/2026 13:58 <DIR> metasploitable3-win2k8
0 fichier(s) 0 octets
3 Rép(s) 111 623 397 376 octets libres

C:\Users\hp\.\vagrant.d\boxes>vagrant init metasploitable3-win2k8
A 'Vagrantfile' has been placed in this directory. You are now
ready to 'vagrant up' your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
'vagrantup.com' for more information on using Vagrant.

C:\Users\hp\.\vagrant.d\boxes>vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Importing base box 'metasploitable3-win2k8'...
==> default: Matching MAC address for NAT networking...
==> default: Checking if box 'metasploitable3-win2k8' version '0.1.0-weekly' is up to date...
==> default: Setting the name of the VM: boxes-default-1788010233170-74348

```

Verified the Metasploitable 3 network configuration with Windows ipconfig; the VM was assigned IPv4 address 172.30.1.22/24.



Verified connectivity between Kali Linux (172.30.1.20) and Metasploitable 3 (172.30.1.22) using ICMP ping. Successful replies confirmed that the VM was reachable on the lab network.

```
(chayma@kali)-[~]
$ ping 172.30.1.22
PING 172.30.1.22 (172.30.1.22) 56(84) bytes of data.
64 bytes from 172.30.1.22: icmp_seq=1 ttl=128 time=1.22 ms
64 bytes from 172.30.1.22: icmp_seq=2 ttl=128 time=0.813 ms
64 bytes from 172.30.1.22: icmp_seq=3 ttl=128 time=0.869 ms
64 bytes from 172.30.1.22: icmp_seq=4 ttl=128 time=1.27 ms
64 bytes from 172.30.1.22: icmp_seq=5 ttl=128 time=1.98 ms
64 bytes from 172.30.1.22: icmp_seq=6 ttl=128 time=1.39 ms
^C
— 172.30.1.22 ping statistics —
6 packets transmitted, 6 received, 0% packet loss, time 5005ms
rtt min/avg/max/mdev = 0.813/1.257/1.981/0.385 ms

(chayma@kali)-[~]
```

- Deployed an Ubuntu Server VM for Snort and verified the Snort installation and version from the command line.

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
chayma@chsnort:~$ snort -V

  ____  _
 / ___|| | | |
| |___| |_| |
 \___|_____|_|_|

-*> Snort! <*-
Version 2.9.15.1 GRE (Build 15125)
By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
Copyright (C) 2014-2019 Cisco and/or its affiliates. All rights reserved.
Copyright (C) 1998-2013 Sourcefire, Inc., et al.
Using libpcap version 1.10.1 (with TPACKET_V3)
Using PCRE version: 8.39 2016-06-14
Using ZLIB version: 1.2.11
```

- Configured the Snort HOME_NET variable for the lab network (172.30.1.0/24), defining the internal network that Snort should monitor.

```
#####
# Step #1: Set the network variables. For more information, see README.variables
#####

# Setup the network addresses you are protecting
#
# Note to Debian users: this value is overridden when starting
# up the Snort daemon through the init.d script by the
# value of DEBIAN_SNORT_HOME_NET s defined in the
# /etc/snort/snort.debian.conf configuration file
#
ipvar HOME_NET 172.30.1.0/24

# Set up the external network addresses. Leave as "any" in most situations
ipvar EXTERNAL_NET any
```

- Created a custom Snort ICMP rule in local.rules to generate an alert when ICMP traffic from any source is directed to the defined HOME_NET.

```
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Detected!!!";sid:1000001;rev:1)
```

Launched Snort in console alert mode and generated ICMP traffic from Kali Linux to Metasploitable 2. Snort detected the traffic and displayed the custom “ICMP Detected!!!” alert in real time, confirming that the rule was triggered successfully.

The image shows two terminal windows. The top window is titled 'Metasploitable2 [En fonction] - Oracle VirtualBox' and shows the output of the 'ip a' command, displaying network interfaces like 'lo' and 'eth0'. The bottom window is titled 'Kali Linux [En fonction] - Oracle VirtualBox' and shows a terminal session where a ping command is executed: 'ping 172.30.1.21'. The output shows five successful ping requests with varying times.

- Added a second custom Snort ICMP rule using dsize:>500 to detect ICMP packets with a payload larger than 500 bytes and generate a “Large ICMP Detected!!!” alert (SID 1000002).

```
chayma@chsnort: ~
GNU nano 6.2 /etc/snort/rules/local.rules
# $Id: local.rules,v 1.11 2004/07/23 20:15:44 bmc Exp $
# -----
# LOCAL RULES
# -----
# This file intentionally does not come with signatures. Put your local
# additions here.
alert icmp any any -> $HOME_NET any (msg:"ICMP Detected!!!";sid:1000001;rev:1;)
alert icmp any any -> $HOME_NET any (msg:"Large ICMP Detected!!!";dsiz>500;sid:1000002;rev:1;)
```

- Tested the oversized-payload rule by sending 501-byte ICMP payloads from Kali Linux to Metasploitable 2 with ping 172.30.1.21 -s 501. Snort detected the traffic and generated the expected “Large ICMP Detected!!!” alert, confirming that the dsiz condition worked.


```

Kali Linux [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
chayma@kali: ~
Session Actions Edit View Help
(chayma@kali)-[~]
$ nmap -sS 172.30.1.21

Starting Nmap 7.98 ( https://nmap.org ) at 2026-08-30 16:09 -0400
Nmap scan report for 172.30.1.21
Host is up (0.0016s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
23/tcp    open  telnet
25/tcp    open  smtp
53/tcp    open  domain
80/tcp    open  http
111/tcp   open  rpcbind
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
512/tcp   open  exec
513/tcp   open  login
514/tcp   open  shell

Selection chayma@chsnort: ~
08/30-20:05:19.004311 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:45434 -> 172.30.1.21:22
08/30-20:05:19.005403 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:45434 -> 172.30.1.21:22
08/30-20:05:19.133648 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:45434 -> 172.30.1.21:22
08/30-20:05:19.140815 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:45434 -> 172.30.1.21:22
08/30-20:05:19.145728 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:45434 -> 172.30.1.21:22
^08/30-20:09:58.340239 [**] [1:1000003:1] FTP connection detected [**] [Priority: 0] {TCP} 172.30.1.20:39219 -> 172.30.1.21:21
08/30-20:09:58.342932 [**] [1:1000003:1] FTP connection detected [**] [Priority: 0] {TCP} 172.30.1.20:39219 -> 172.30.1.21:21
08/30-20:09:58.346366 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:39219 -> 172.30.1.21:22
08/30-20:09:58.349866 [**] [1:1000004:1] SSH connection detected [**] [Priority: 0] {TCP} 172.30.1.20:39219 -> 172.30.1.21:22
08/30-20:09:58.360522 [**] [1:1418:11] SNMP request tcp [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 172.30.1.20:39219 -> 172.30.1.21:22
08/30-20:09:58.427434 [**] [1:1421:11] SNMP AgentX/tcp request [**] [Classification: Attempted Information Leak] [Priority: 2] {TCP} 172.30.1.20:39219 -> 172.30.1.21:22

```

- Tested the custom HTTP rule by requesting `http://172.30.1.21/chaymaadmin.php` from Kali Linux with curl. The web server returned 404 Not Found, but the requested URL matched the Snort rule and generated the “Attempt to get into protected page” alert (SID 1000005).

```

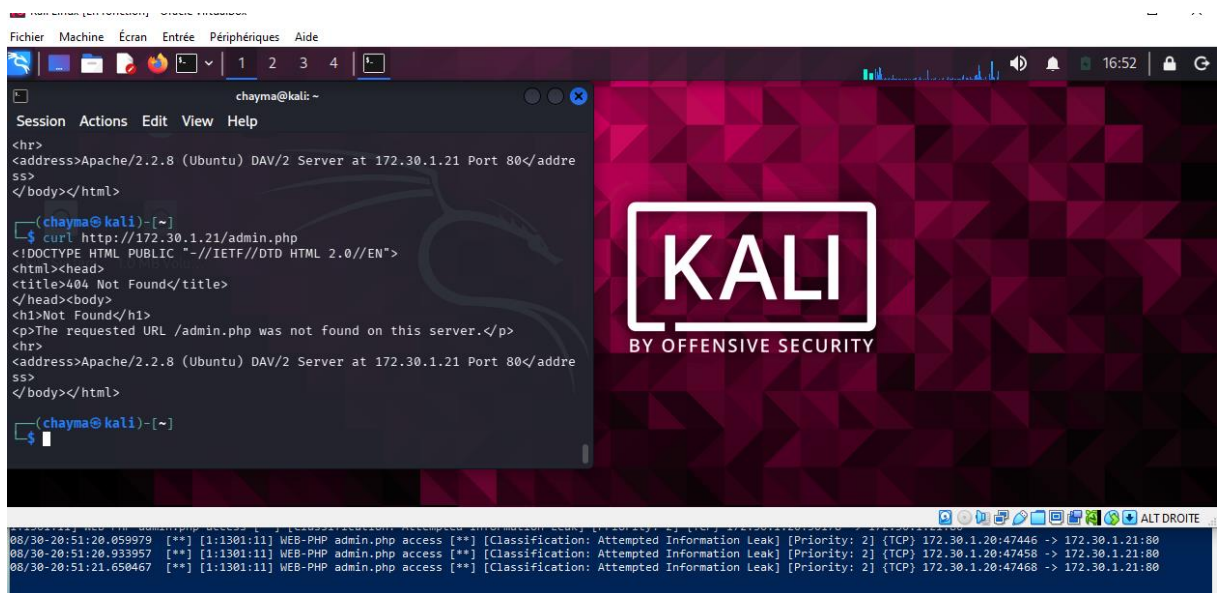
(chayma@kali)-[~]
$ curl http://172.30.1.21/chaymaadmin.php
<!DOCTYPE HTML PUBLIC "-//IETF//DTD HTML 2.0//EN">
<html><head>
<title>404 Not Found</title>
</head><body>
<h1>Not Found</h1>
<p>The requested URL /chaymaadmin.php was not found on this server.</p>
<hr>
<address>Apache/2.2.8 (Ubuntu) DAV/2 Server at 172.30.1.21 Port 80</address>
</body></html>

chayma@chsnort: ~
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

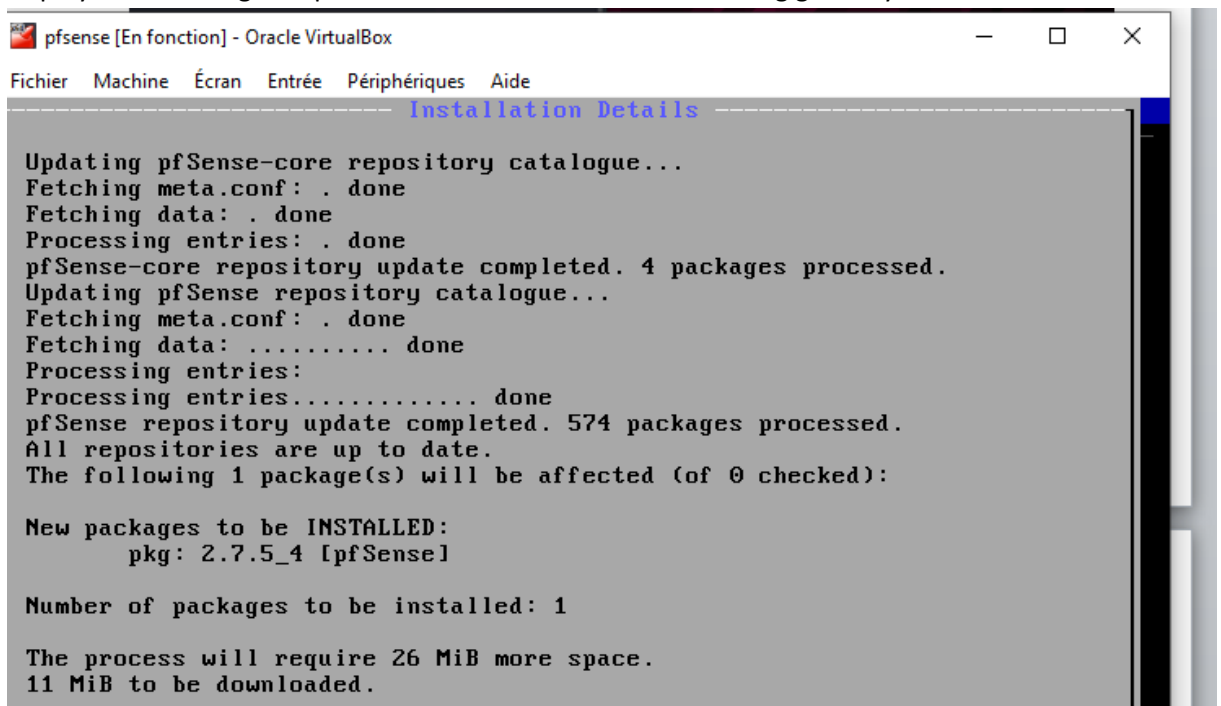
Last login: Sun Aug 30 18:47:09 2026 from 192.168.43.61
chayma@chsnort: $ sudo snort -q -l /var/log/snort -i enp0s8 -A console -c /etc/snort/snort.conf
[sudo] password for chayma:
08/30-20:42:12.094435 [**] [1:1000005:1] Attempt to get into protected page [**] [Priority: 0] {TCP} 172.30.1.20:37724 -> 172.30.1.21:80

```

- Tested Snort’s built-in detection rules by requesting `http://172.30.1.21/admin.php` from Kali Linux. The request triggered the pre-installed community rules and generated a WEB-PHP admin.php access alert (SID 1301), classified as an attempted information leak.



- Deployed and configured pfSense as the lab's firewall and routing gateway.



- Configured the pfSense LAN interface (em1) with the static IPv4 address 172.30.2.1/24 and disabled DHCP on that interface. This address serves as the default gateway for hosts on the internal lab network.


```

4) Reset to factory defaults      13) Update from console
5) Reboot system                  14) Enable Secure Shell (sshd)
6) Halt system                    15) Restore recent configuration
7) Ping host                      16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Configure IPv4 address LAN interface via DHCP? (y/n) n

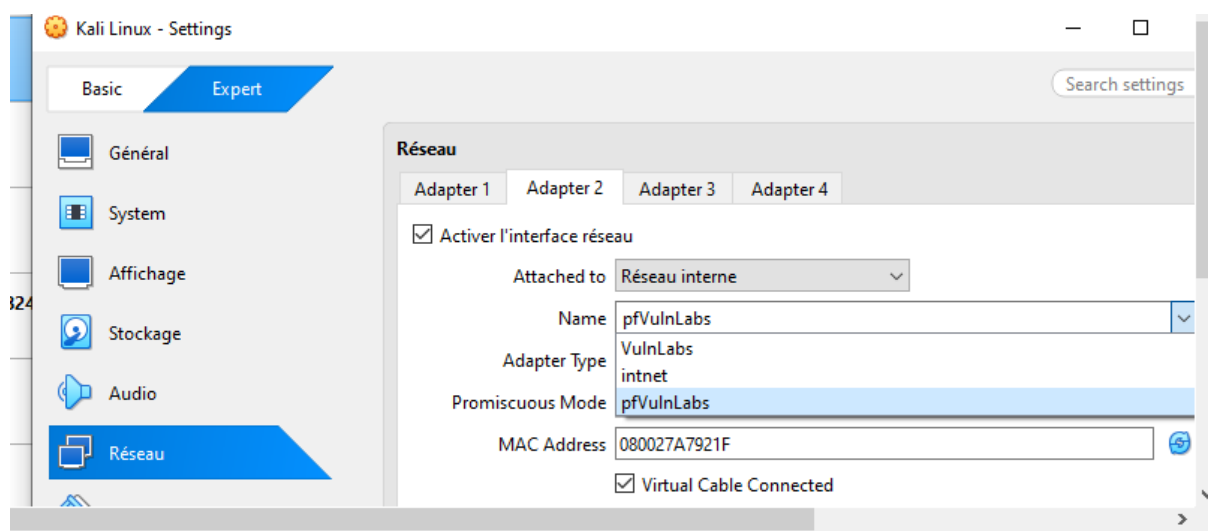
Enter the new LAN IPv4 address. Press <ENTER> for none:
> 172.30.2.1/24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

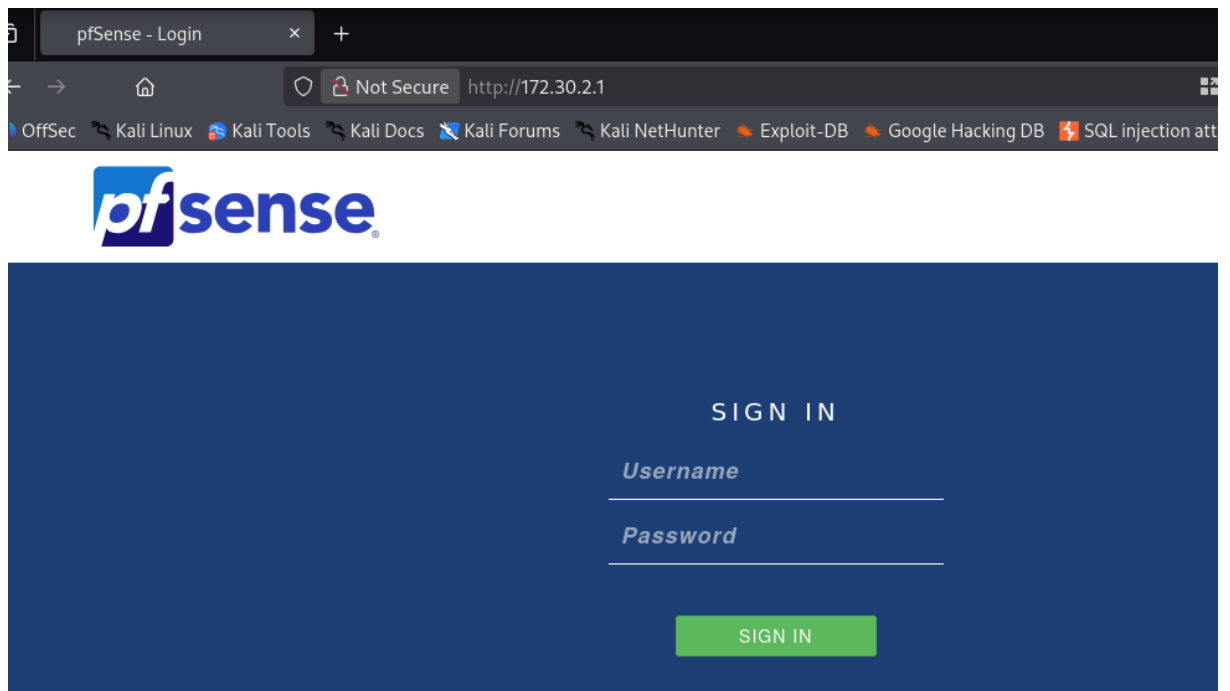
Configure IPv6 address LAN interface via DHCP6? (y/n)

```

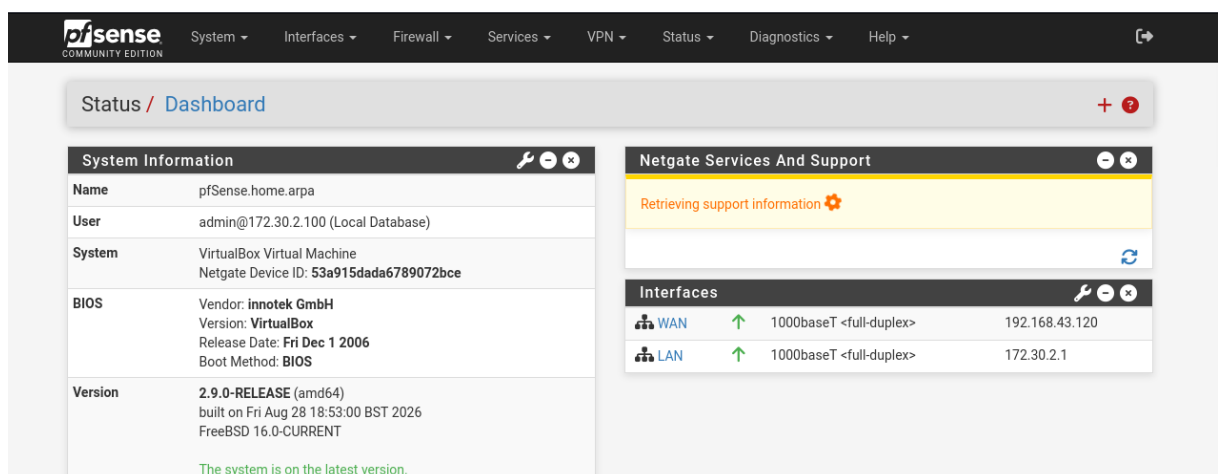
Connected Kali Linux, Metasploitable 2, and Metasploitable 3 to the pfVulnLabs internal network in VirtualBox. This placed the three VMs on the isolated lab segment behind the pfSense LAN interface.



- Verified LAN connectivity to pfSense by accessing its WebGUI at <http://172.30.2.1> from Kali Linux. The successful login page confirmed that Kali could reach the pfSense LAN interface over the internal lab network.



- Verified the pfSense dashboard and interface status. The WAN interface was shown at 192.168.43.120 and the LAN interface at 172.30.2.1, confirming that pfSense had both an upstream connection and an active internal LAN interface for routing and firewall functions.



- Verified outbound routing through pfSense by pinging google.com from Metasploitable 2. Successful replies confirmed that internal traffic could be routed through the pfSense LAN and WAN interfaces to the external network.

```
Metasploitable2 [En fonction] - Oracle VirtualBox
Fichier Machine Écran Entrée Périphériques Aide
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast qlen 1000
    link/ether 08:00:27:16:d7:4c brd ff:ff:ff:ff:ff:ff
    inet 172.30.2.101/24 brd 172.30.2.255 scope global eth0
    inet6 fe80::a00:27ff:fe16:d74c/64 scope link
        valid_lft forever preferred_lft forever
msfadmin@metasploitable:~$
msfadmin@metasploitable:~$ ping google.com
PING google.com (142.251.27.138) 56(84) bytes of data:
64 bytes from cv-in-f138.1e100.net (142.251.27.138): icmp_seq=1 ttl=103 time=103.
9 ms
64 bytes from cv-in-f138.1e100.net (142.251.27.138): icmp_seq=2 ttl=103 time=51.
9 ms
64 bytes from cv-in-f138.1e100.net (142.251.27.138): icmp_seq=3 ttl=103 time=45.
9 ms
64 bytes from cv-in-f138.1e100.net (142.251.27.138): icmp_seq=4 ttl=103 time=63.
1 ms
64 bytes from cv-in-f138.1e100.net (142.251.27.138): icmp_seq=5 ttl=103 time=60.
6 ms
64 bytes from cv-in-f138.1e100.net (142.251.27.138): icmp_seq=6 ttl=103 time=66.
2 ms

--- google.com ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 6025ms
rtt min/avg/max/mdev = 45.934/65.277/103.728/18.521 ms
msfadmin@metasploitable:~$
```

- Created a pfSense LAN firewall rule with Action set to “Block”, Address Family set to IPv4, Protocol set to Any, and Source set to LAN subnets.
Set the rule Destination to Any, causing matching IPv4 traffic originating from the LAN subnet to be blocked by the firewall.

Edit Firewall Rule	
Action	Block Choose what to do with packets that match the criteria specified below. Hint: the difference between block and reject is that with reject, a packet (TCP RST) whereas with block the packet is dropped silently. In either case, the original packet
Disabled	☐ Disable this rule Set this option to disable this rule without removing it from the list.
Interface	LAN Choose the interface from which packets must come to match this rule.
Address Family	IPv4 Select the Internet Protocol version this rule applies to.
Protocol	Any Choose which IP protocol this rule should match.

Source

Source ☐ Invert match LAN subnets Source Address /

Destination

Destination ☐ Invert match Any Destination Address /

Extra Options

- Tested the blocking rule by pinging google.com again from Metasploitable 2. No replies were received, demonstrating that the new LAN firewall rule blocked the outbound traffic.

```
msfadmin@metasploitable:~$ ping google.com
```

- Configured a pfSense WAN port-forwarding rule to redirect selected inbound connections received on the WAN address to Metasploitable 2 at 172.30.2.101. This demonstrates how pfSense can publish a selected internal service through the WAN interface using destination NAT/port forwarding.

Not Secure http://172.30.2.1/firewall_nat_edit.php?id=0

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB SQL injection attack, q...

No RDR (NOT) ☐ Disable redirection for traffic matching this rule
This option is rarely needed. Don't use this without thorough knowledge of the implications.

Interface WAN
Choose which interface this rule applies to. In most cases "WAN" is specified.

Address Family IPv4
Select the Internet Protocol version this rule applies to.

Protocol Any
Choose which protocol this rule should match. In most cases "TCP" is specified.

Source [Display Advanced](#)

Destination ☐ Invert match. WAN address
Type Address/mask

Redirect target IP Address or Alias 172.30.2.101
Type Address